

The Path Back to Low Inflation in New Zealand

Delivery: 1.15pm, 23 March 2023

Venue: Intercontinental Hotel, Wellington



Reserve Bank
of New Zealand
Te Pūtea Matua

Introduction

Thank you for inviting me to speak today. It's a pleasure to be here, especially in person.

I'm going to talk about why we are currently suffering a serious bout of inflation and what the Monetary Policy Committee (MPC) at the Reserve Bank of New Zealand is doing about it. This is important not only for the Reserve Bank, but for all New Zealanders because inflation impacts all of us.

According to the latest data – which is for the December quarter last year – annual consumer price inflation was 7.2%, which is around the highest it has been since the late 1980s (Figure 1).

High inflation creates economic costs and distortions.¹ For example, unexpected inflation makes it more difficult for people to see relative price changes in the economy, increasing the likelihood of poor decisions being made. People devote more time and effort to minimising inflation risks, rather than more productive endeavours.

Even when expected, inflation still hurts. For example, when interest rates increase to compensate for the declining value of money, lenders pay tax on the additional interest and borrowers may have to cut other spending to pay higher interest bills.

Overall, inflation leads to misallocated resources and lower living standards. Less wealthy and less financially sophisticated people are more likely to lose out.

Keeping inflation low and stable, as was generally the case in the three decades prior to the COVID-19 pandemic, is the best contribution monetary policy can make to promoting economic wellbeing and prosperity. This is enshrined in the Reserve Bank of New Zealand Act (2021) and why maintaining price stability is a fundamental part of the MPC's Remit.²

We remain deeply committed in our work of bringing inflation down and keeping it low and stable in future. Quite simply, that's our job.

I want to leave you with four key messages from this speech:

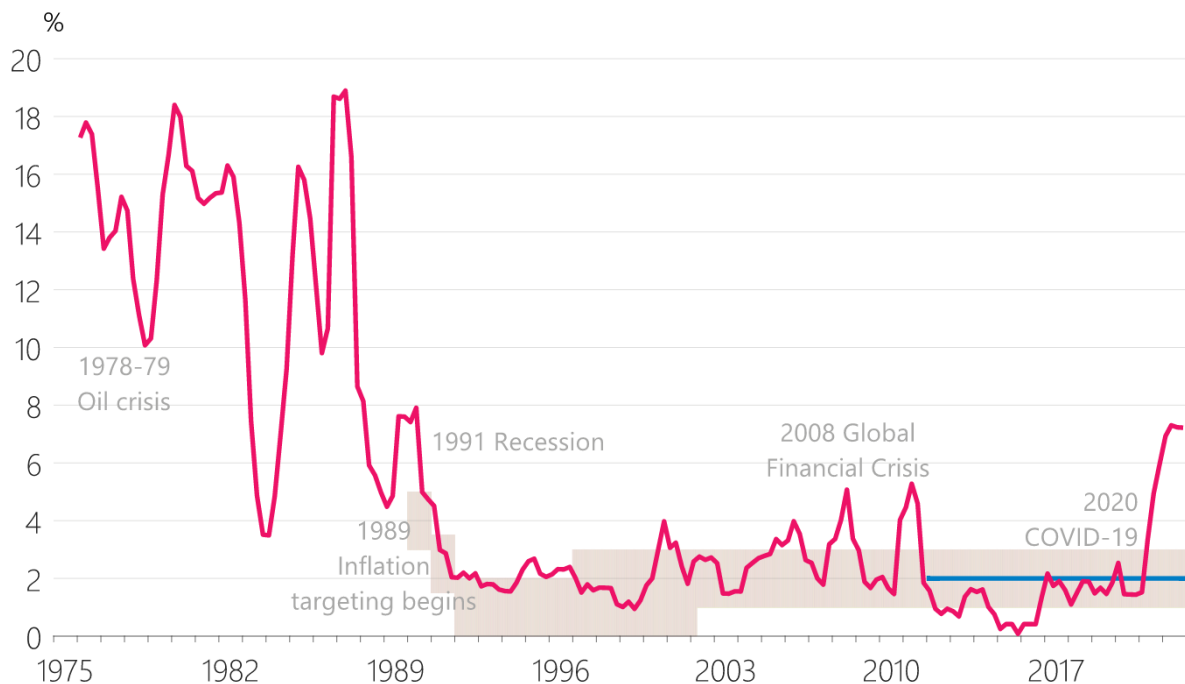
- **Inflation is high and widespread** because strong demand outstripped supply. Businesses passed on higher costs and workers sought higher pay to compensate for higher prices. Fiscal and monetary responses to the pandemic supported demand while health measures and other shocks reduced the supply of labour and products.
- **We are worse off because of the pandemic, the war and floods.** Monetary policy cannot do anything about the loss of real income stemming from these events.
- **Monetary policy is lowering inflation** by lifting interest rates to ensure persistence in inflation above our inflation target is squeezed out of the economy. We are incredibly determined to get inflation and inflation expectations back to target.
- **Returning inflation to target could be made more difficult** if businesses and workers try to push up their real profit margins and real wages to make up for the inflationary impact of the

¹ See, for example, Samantha Johnson (1993) "The costs of inflation revisited", *Reserve Bank Bulletin*, Vol 56, No 1. Pp 3-31.

² Reserve Bank of New Zealand Act 2021 (legislation.govt.nz/act/public/2021/0031/latest/LMS286978.html) and Reserve Bank of New Zealand (Replacement of Remit for Monetary Policy Committee) Order 2021 Reserve Bank of New Zealand (Replacement of Remit for Monetary Policy Committee) Order 2021 (LI 2021/29) Contents – New Zealand Legislation.

pandemic, the war, and the storms. In this case, Monetary Policy would need to be more contractionary for longer, resulting in a deeper recession.

Figure 1: Annual CPI inflation in New Zealand



Source: Stats NZ, RBNZ estimates.

Note: Shaded bar represents the inflation target range since the introduction of inflation targeting in 1989. From 2012, a focus on the 2 per cent target midpoint was added (blue line).

How Did We Get High Inflation?

Reduced Ability to Produce Goods and Services

The COVID-19 pandemic and war and related events reduced the ability of the global economy to produce goods and services. Global supply chains were severely disrupted, especially for us living here in the South Pacific – the last bus stop on the planet. This really was a severe “black swan event” for global logistics.

Labour shortages have also been incredibly intense globally and in New Zealand. As COVID-19 spread, a large share of the global workforce was sick or isolating and couldn't work. For example, in March 2022, the number of people off work in New Zealand for a week or more because of sickness was up almost 70% compared with a year earlier.³

It is not just the health impacts of COVID-19 that led to worker shortages. In some countries, the pandemic caused people to reassess their priorities and, in many cases, to stop working for a time or retire early.

We didn't see so much of this in New Zealand, where participation in the labour market remained incredibly strong and is currently at record highs. But when our border closed in the early days of the pandemic, international arrivals and departures collapsed to near zero. Net migration then

³ “More people away from work due to sickness” – Stats NZ (stats.govt.nz/news/more-people-away-from-work-due-to-sickness/).

turned negative through 2020 and 2021 as permanent or long term departures increased while arrivals stayed low. This was in stark contrast to previous years of very large migration inflows.

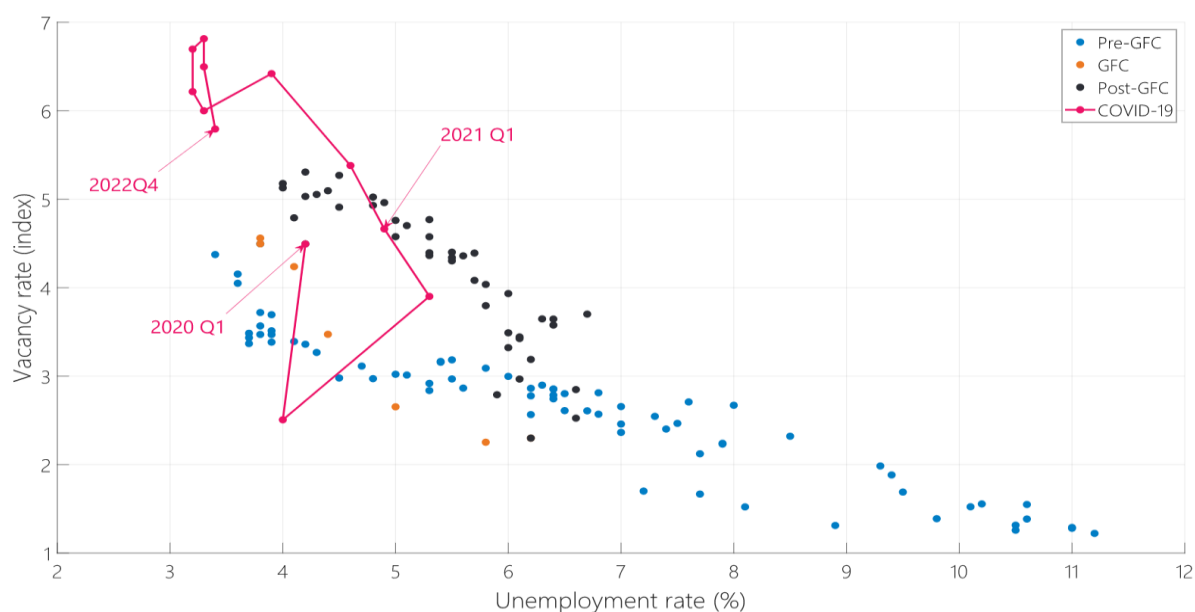
Our labour market tightened. Over mid-2020, suggestive of structural change in the economy, unemployment and job vacancies both increased (Figure 2). But from the end of 2020, unemployment fell and job vacancies increased further as workers became increasingly difficult to find. The share of people moving between jobs also increased strongly, as competition for workers increased (Figure 3).

As borders re-opened, younger New Zealanders have been leaving, further reducing the domestic pool of workers. The labour market has remained incredibly tight, with high vacancy rates and low unemployment.

The recent summer flooding in Auckland and Cyclone Gabrielle have also reduced supply in some industries.

You will have no doubt experienced the results of supply chain disruptions and worker shortages over recent years: for example, shops being closed because of staff illness; long waits for online purchases as international and domestic supply chains came under pressure; high food price inflation, and high petrol prices when the war in Ukraine impacted energy markets.

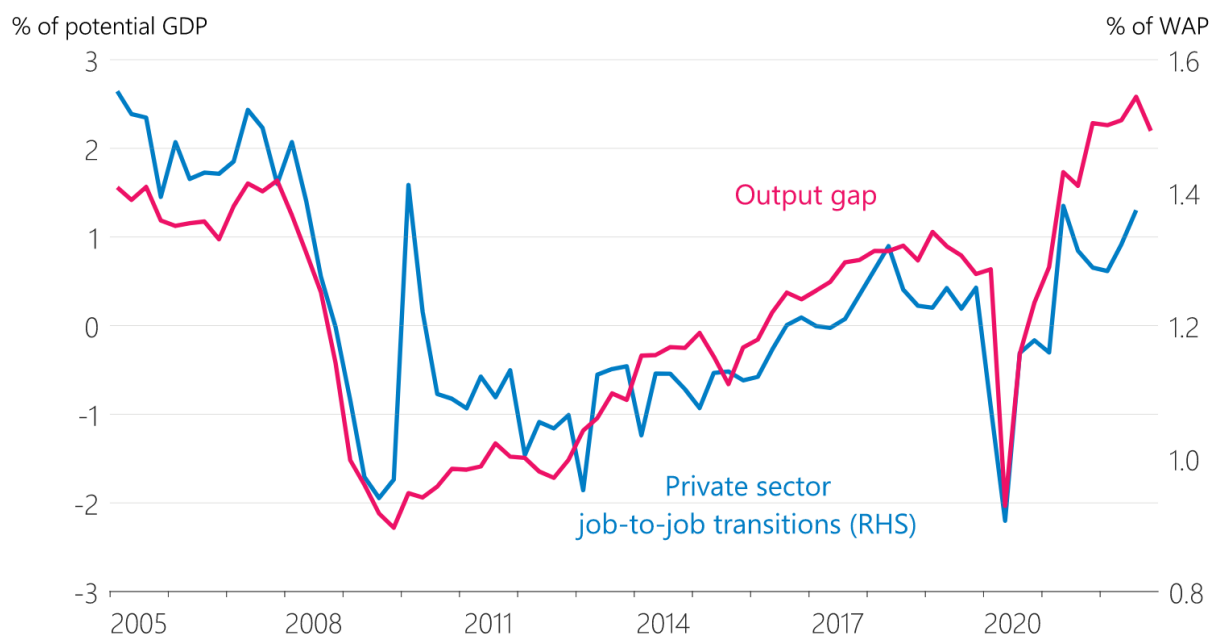
Figure 2: Unemployment and the job vacancy rate



Source: Stats NZ, Ministry of Business, Innovation and Employment (MBIE), RBNZ estimates.

Note: The vacancy rate is defined as the number of job vacancies as a share of the labour force. Colours reflect different periods in recent history. Pink: COVID-19 (2020-2022); Black: post-GFC (2009Q3-2019); Orange: GFC (2008-2009Q2); blue: pre-GFC (1990-2007).

Figure 3: Job-to-job transitions



Source: Stats NZ, Stats NZ IDI, RBNZ estimates.

Note: WAP stands for 'working-age population'. Private sector job-to-job transitions exclude transitions in the Education and Training and Public Administration and Safety industries. Job-to-job transitions data points for December 2017 and March 2018 have been removed due to data quality concerns.

Resilient Demand and Spending

While the supply of goods and services was restricted, spending and demand in the economy remained strong. Quite simply, demand outstripped our ability to supply.

For us in New Zealand, this partly reflected the effectiveness of our health response to COVID-19. While lockdowns were tough going – especially for people living in Auckland – economic life for many New Zealanders was surprisingly 'normal' for many months during the global pandemic.

At the same time, Government support to the economy increased over 2020. New packages, such as the Wage Subsidy Scheme, were designed and rolled out at pace. Interest rates were pushed lower, with the Official Cash Rate (OCR) dropped to its 'effective lower bound' and additional monetary policy tools put in place.

For the most part, people remained in work, which supported income growth. There were also large build-ups of cash balances and savings, given limited spending opportunities during lockdowns. This also supported household spending and demand in the economy as restrictions were gradually eased.

Demand has been resilient overall, but there have been big swings in what people have been spending their money on, given changing consumer preferences and economic realities. Perhaps most obviously, as borders around the world closed, a precipitous drop in visitor arrivals hit tourism spending hard.

More generally, spending requiring face-to-face contact – mostly in service industries such as cafes, bars, and shops – fell sharply as the pandemic hit. On the flip side, spending on goods that could be bought online and delivered surged. The online economy grew strongly over this period.

Construction also boomed as resilient incomes, low unemployment, and low interest rates increased demand for housing. People spending more time at home – either working from home or locked down – also increased housing demand.

With constraints on housing supply, stronger demand contributed to a 44% increase in house prices from early 2020 to late 2021.⁴ Unusually, given we were in the middle of a global pandemic, household wealth increased, adding to spending pressures *via* a wealth effect. That is, people tend to spend more when their wealth increases.

More recently, as health restrictions were progressively removed, demand for services has increased strongly, particularly in hospitality and tourism. Demand for used cars and other products destroyed in the cyclone has also spiked.

So, over the last few years, the story has been one of resilient demand but with big swings and changes in its composition.

Resilient Demand at a Time of Disrupted and Limited Supply Causes Inflation

Resilient demand plus supply-side issues equals inflation. To start with, bottlenecks led to sharp price increases for specific products. Perhaps the most obvious example was when supply restrictions caused by the war in Ukraine sent global energy and food prices sharply higher.

As a small open economy, it was impossible for New Zealand to avoid inflationary pressures caused by global events. Almost half of headline inflation in New Zealand over the pandemic originated offshore (Figure 4).

Keeping headline inflation in the 1-3% target over this time would have required monetary policy to be highly contractionary to reduce home-grown (non-tradable) inflation, in order to offset imported inflationary pressure.

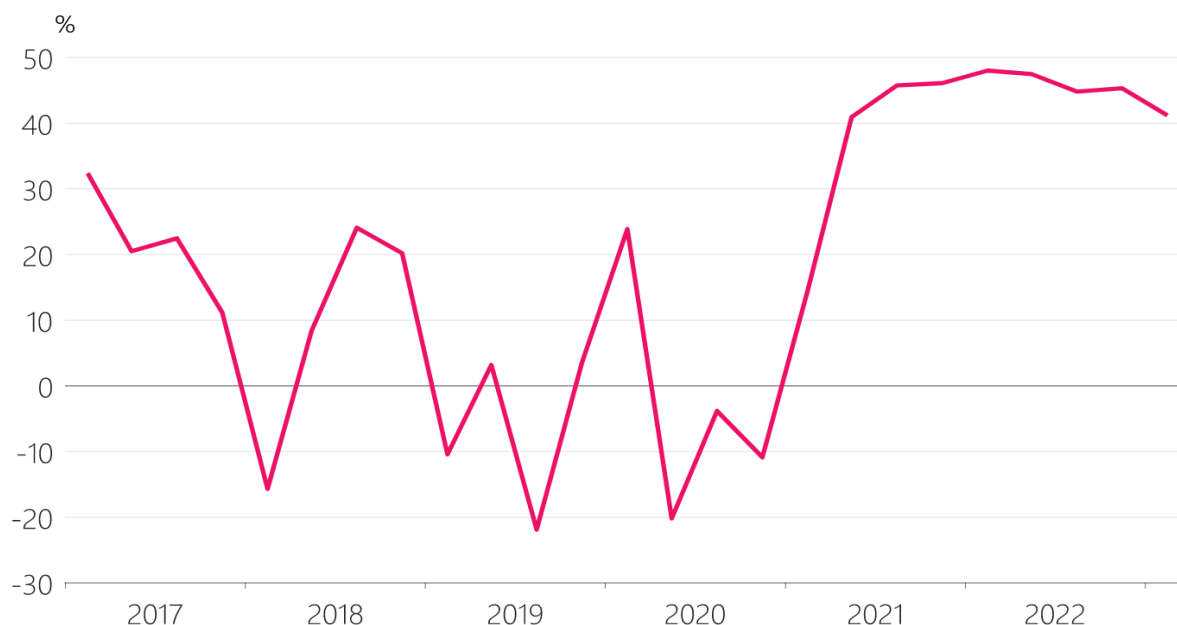
Excess demand dynamics also played out in some domestic markets. For instance, construction and housing-related costs are less influenced by global events but have risen sharply over the past two years because of difficulties finding materials (such as Gib plasterboard) and workers.

Inflation in the tourism industry is another example. Tourism is on the front line of labour shortages and still rebuilding capacity lost during the early years of the pandemic. With higher demand driven by a rapid rebound in international visitors, prices charged by many tourism-related businesses have increased sharply.

⁴ Housing (Still) Matters – The Big Picture rbnz.govt.nz/hub/publications/speech/2022/speech2022-06-30

Big swings in the composition of demand may also be inflationary. Recent work from the Bank of England finds that prices don't fall much in industries hit with weaker demand but increase quickly in industries facing into stronger demand.⁵ This asymmetric response might explain why overall inflation did not fall in 2020 but increased strongly as demand started rebounding.

Figure 4: Tradable inflation as a share of headline inflation



Source: Stats NZ

Inflation Becomes Widespread and Persistent

In 2021, inflation increased mainly because of a relatively small number of large price increases, suggesting excess demand for a few specific products (Figure 5). In 2022, small price increases across a large number of products contributed to inflation, suggesting widespread inflation pressures in the economy.

Inflation can become widespread in a number of ways. Price increases may spread through the economy as businesses try to pass on higher input costs and workers try to maintain the spending power of their incomes by seeking higher pay.

With a very tight labour market and weak competition in some parts of our economy, the scope for price increases to become embedded in cost-price dynamics and trigger second-round inflationary effects is high. These second-round effects create ongoing inflation, even after the original stimulus – a relative price shock for specific products like petrol or plasterboard – has died away.

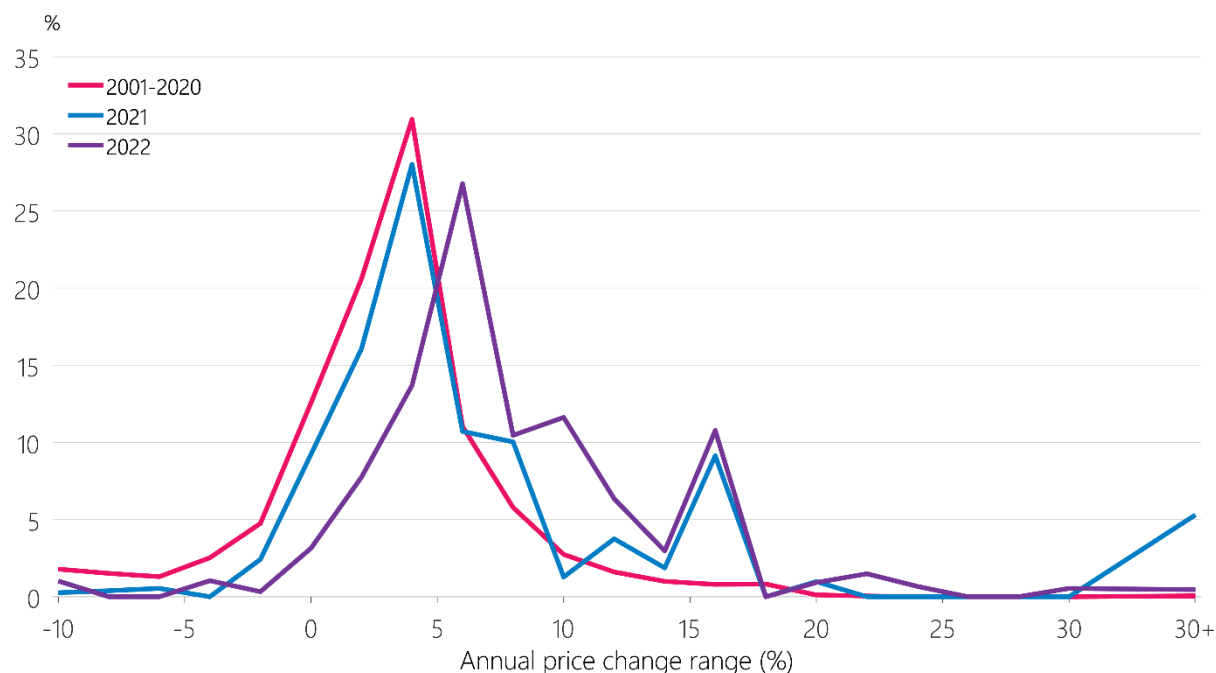
Inflation can also spread through the economy because it is easier for firms to increase prices when other firms are doing likewise. Customers may think it more reasonable that firms increase their prices when there is inflation in the economy.

Inflation can become persistent if people expect it to stay high in future. If people make decisions today based on the expectation of higher prices in future, then inflation will persist. Inflation

⁵ Bunn, P., Anayi, L. S., Bloom, N., Mizen, P., Thwaites, G., & Yotzov, I. (2022). Firming up price inflation (No. w30505). National Bureau of Economic Research.

expectations can become self-fulfilling in that once inflation starts, the belief that it will continue can sustain it.

Figure 5: The distribution of price changes



Source: RBNZ calculations using Statistics New Zealand CPI level 3 class data, INFOS CPI013AA, containing price indices for 109 categories (e.g. Bread and cereals; Women's clothing; Postal services).

Note: The graphs show the distribution of annual price increases each year, calculated as the fraction of items which have a price increase in 2 percentage point intervals eg 0 - 2%, 2 - 4%. The fractions are calculated using contemporaneous CPI weights. For example, between 2001 and 2020, 31% of categories had price increases between 2 and 4%, on average; this fraction fell to 28% in 2021 and 14% in 2022. The graph shows that in 2021 there was only a small reduction in the number of categories having price increases of 4% or less, but a large increase in the number of categories having very large price increases, particularly price increases above 30%. In contrast, in 2022 there was a rightward shift of the whole distribution, with a large reduction in the number of categories with less than 4% price increases, and a large increase in the number price increases between 4 and 12%.

How to Tame Inflation

What Monetary Policy Can't Do

Before outlining how MPC is working to lower inflation, I want to briefly touch on what monetary policy can't do. Monetary policy can't do anything about income lost because of the pandemic, the war in Ukraine, and recent natural disasters.

These losses reflect a reduced ability to produce goods and services. While monetary policy can and must respond to inflationary pressures caused by excess demand, it cannot do much to improve the ability of our economy to produce output. For example, the Reserve Bank can do nothing to make up for a shortage of kumara or onions or apples caused by flooding.

Instead, improving productivity is the best way to lift incomes sustainably. By alleviating supply constraints, better productivity would also reduce long-run inflationary pressures at minimal economic cost. More productive businesses can hold or even cut prices while paying workers more and remaining competitive.

True, lifting productivity has not been easy in New Zealand, which is why New Zealanders work more hours than most in the developed world but earn relatively low incomes.

One reason for our poor productivity is that competition may be weak in some New Zealand markets, making it easier for businesses to pass on cost increases to their customers.

Improving competition is about regulatory changes that lower barriers to entry and discourage anti-competitive behaviours. It is excellent to see the Commerce Commission now doing Market Studies and having better regulatory tools at its disposal to pursue abuse of dominance cases.

There are also things people can do to enhance the power of competition and keep price pressures in check. Shop around for that good deal. Wait for the sales. Use price comparison websites. And go to the effort of changing banks or utility providers if it means getting a better deal.

How MPC is Working to Lower Inflation

While higher productivity and more competitive markets are essential for lifting our economic performance and blunting inflationary pressures over the long run, the responsibility for reducing current inflation back into the 1-3% target band sits firmly with MPC.

With inflation above 7% and far outside the target range, MPC has been increasing the OCR to bring demand back into balance with the supply capacity of our economy.

Ideally, and with the benefit of hindsight, monetary policy should have started tightening earlier in 2021. This, along with many other lessons learned, was recognised by the Reserve Bank in our recent review of monetary policy over the past five years.⁶

In any case, at 4.75%, the OCR has increased by 450 basis points in only 18 months. In nominal terms, this is the most rapid increase in its history and the highest it has been since before the 2008 Global Financial Crisis.

Higher Interest Rates Soften Demand and Lower Inflation

In short, monetary policy shifts demand in the economy through time. Higher interest rates reduce demand by encouraging people to spend less and save more while lower interest rates have the opposite effect. Higher interest rates push economic activity into the future whereas lower interest rates pull activity into the present.

⁶ In retrospect: Monetary policy in New Zealand 2017 to 2022 – bnz.govt.nz/hub/publications/monetary-policy-statement/rafimp.

So, as we work to rein in inflation, mortgage-holders and businesses looking for finance are seeing their interest rates increasing. People with savings are seeing the nominal return on their savings increase. However, the after-tax real return on savings may still be negative, given high inflation, sluggish increases in terms deposit rates, and tax settings.

But over time, higher interest rates will encourage saving and restrict spending by households with mortgages. They will also hold back hiring decisions, and may lead to lower cash flow and lower asset prices, as we have already seen with house prices. Less helpfully, higher interest rates may also restrict business investment.

All that will help lower aggregate demand in the economy. In turn, this will mean less excess demand in product and labour markets, which will mean lower inflation and employment moving back towards its maximum sustainable level.

As in many countries around the world, we expect this monetary policy tightening to cause the New Zealand economy to enter a mild recession later this year as demand slows.

Expectations Really Matter

The effectiveness of higher interest rates in bringing balance to the economy and lowering inflation depends on how realistic and pragmatic New Zealanders are in accepting that we are worse off than we otherwise would be given global events of the past few years and recent storms.

As Governor Adrian Orr puts it, the people of planet earth are poorer because of the pandemic and war. The recent floods in Auckland and the cyclone along the East Coast have also reduced wealth and prosperity in New Zealand. In practice, this means accepting the fact of higher prices for some producer inputs and consumption goods, meaning lower real wages and profits than otherwise.

If businesses and workers try to push their real profit margins and real wages to where they would have been if not for the pandemic, the war, and storms, then the more likely it is that high inflationary pressures persist into the future.

Resisting the squeeze on real incomes and spending power would mean MPC has to increase interest rates further or hold them up for longer, and the deeper any recession would need to be to get inflation back within the target band.

As I outlined earlier, monetary policy cannot do anything about the loss of real income stemming from global events and natural disasters. What we can and are doing is lifting interest rates to ensure any self-sustaining persistence in inflation above the inflation target is squeezed out of the economy by constraining demand relative to supply as necessary.

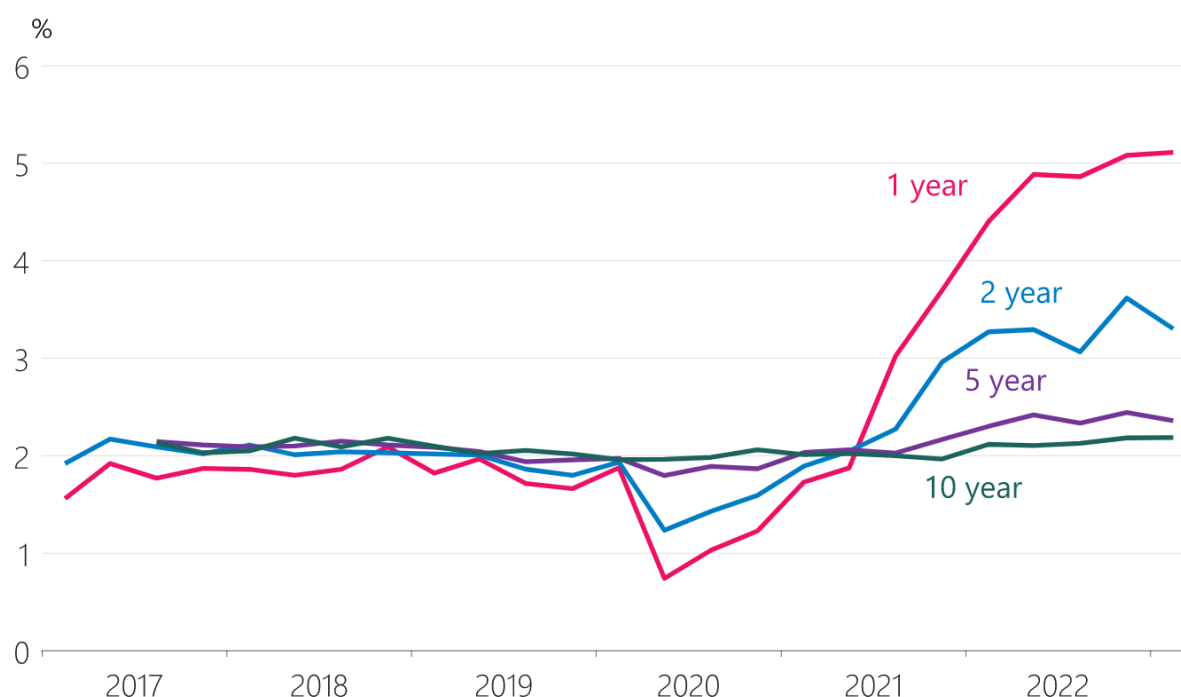
By the same token, if businesses and households make decisions today based on the expectation of rising prices in future, then inflationary pressures will persist. Higher interest rates for longer and a deeper recession would again be necessary to bring inflation back to target.

Conversely, if people expect inflation to be within the 1-3% target in future and act accordingly, the less work we have to do with higher interest rates that weaken demand.

Taming inflation means ensuring that short-term inflation expectations do not just reflect current high inflation and that medium-term inflation expectations do not drift so that long-term expectations remain anchored to the inflation target (Figure 6).

The path back from persistent to low inflation always involves some output losses and lower incomes overall. The size of those losses depends on how realistic we all are in adjusting to new economic realities and in expecting inflation to fall back to target, consistent with MPC's Remit.

Figure 6: Inflation expectations



Source: RBNZ Survey of Expectations (Business).

Monetary Policymaking in a World of Change

If that didn't already sound complicated, let me layer on some of the complexities and uncertainties MPC faces when making interest rate decisions. (And can I stress that decisions on the OCR are reached through consensus by the seven members of MPC.)

First, the global economy is highly volatile, fragile, and uncertain. Economic growth rates have been slowing, partly in response to higher interest rates, and international organisations paint a gloomy future for the global economy.

Globalisation is changing, with big implications for small economies such as ours. The global trading system is fracturing along political lines and geopolitical tensions are rife. The transition to net zero carbon emissions will lead to further changes in the global balance of power and trade relations.

This presents MPC with a very volatile and challenging international environment to navigate over the current cycle. It also raises big questions about the type of economy we will face once inflation is back within the target band.

On the domestic front, our hearts go out to people impacted by the recent storms and floods in the North Island. The material and financial losses are clearly substantial.

With no spare capacity in the economy, diverting resources into cyclone-affected areas for the rebuild will mean reducing activity in other parts of the country. There are a number of ways this rebalancing could occur, each with different implications for interest rate.

If the rebuild generates inflationary pressure, interest rates would increase, reducing demand and freeing up resources for the rebuild. Alternatively, diverting Government resources from projects elsewhere in the country or funding the rebuild *via* increased government revenues would free up resources without the need for higher interest rates.

Another uncertainty growing increasingly pertinent as we near the interest rate peak is knowing when monetary policy has done enough to get inflation comfortably back to target.

The effects of the 450 basis point increase in the OCR over the last 18 months are still percolating through the economy. For instance, over the coming year, 50% of mortgages will roll off low fixed-term rates onto much higher rates. The 'average' mortgage rate will move from 4.5% to over 6%.

We assume this will affect consumer spending in a similar way to interest rate increases in previous cycles. But the rapid speed of rate increases creates additional uncertainty about how the effects of the tightening will play out.

It is also unclear whether inflation expectations are falling fast enough to mean current and projected OCR settings are consistent with positive real interest rates, and thus likely to be disinflationary. In part, this comes down to how convinced people are that MPC will do what's necessary to get inflation down.

On balance, our measures of neutral interest rates – the interest rate that is neither stimulatory nor contractionary – indicate the OCR is now comfortably above neutral and having the desired contractionary effect. Consistent with this, and as discussed in detail in the February *Monetary Policy Statement*, there are welcome signs demand in the economy is slowing.

Achieving the Remit

Monetary policy is not just about returning inflation to target and keeping it there. The Reserve Bank of New Zealand Act (2021) instructs us to also support maximum sustainable employment.

Currently, with inflation well above target and employment clearly above its maximum sustainable level, achieving this dual mandate requires contractionary monetary policy to slow domestic spending and employment and to lower inflation.

However, in a stagflation environment of low growth, rising unemployment and high inflation, the optimal path for interest rates becomes less clear. Would MPC raise interest rates to control inflation or lower rates to encourage growth and jobs?

In this situation, history and economic theory suggest a policy response aimed at a more drawn-out return of inflation to target and a smaller decline in employment.

But if inflation expectations move higher, the optimal response is to put greater weight on returning inflation to target. This would support maximum sustainable employment in the medium

term by reducing the need for an extended period of contractionary monetary policy to get even higher inflation expectations back to target.

Our Monetary Policy Remit also instructs us to have regard to the efficiency and soundness of the financial system and to avoid unnecessary instability in output, interest rates, and the exchange rate. We also have to discount transitory inflationary events and assess the impact of MPC decisions on the Government's policy on house prices.

In practice, this means we have a medium term focus for our price stability objective. The flexibility that allows us is important in that it ensures we can maintain price stability over time without causing unnecessary damage to the economy.

We are currently reviewing our Remit and will give advice to the Minister of Finance in April.

Conclusion

The title of this speech is how to get back to low inflation in New Zealand. In a narrow sense, this is about getting monetary policy right so inflation returns to target. This means firms, workers and taxpayers accepting the distributional consequences of recent global and domestic shocks and lowering their inflation expectations accordingly.

We are well placed to cope with this adjustment. The economy has grown strongly over the pandemic and unemployment and headline CPI are both in the lower third of OCED economies, while core inflation is about middle of the pack. Household balance sheets are in good shape in aggregate and the banking system is very well capitalised, meaning there is no conflict between monetary policy and financial stability.

In a broader sense, getting monetary policy right in a complex and volatile environment means building a deep understanding of the evolving economic context in which monetary policy operates. This requires an ambitious economic research agenda, including the use of new and improved data for monitoring and understanding the economy.

It also requires strong collaboration across the public sector and with the domestic and international academic community. For example, we must work with the Treasury to understand the future role of fiscal policy instruments in managing economic shocks.

In the broadest sense, getting back to low inflation is about improving the ability of the New Zealand economy to thrive in a rapidly evolving global environment. But that's a conversation for another day.